

Splunk: Detecting Beaconsing Malware

Objective

The objective of this lab is to detect beaconsing malware using Splunk.

Steps

1. Beaconsing Malware Detection

Used Splunk to identify and detect instances of beaconsing malware behavior within network traffic or endpoint logs.



The screenshot shows the Splunk search interface with a search query for beaconsing malware. The query is as follows:

```
1 index="cobaltstrike_beacon" sourcetype="bro:http:json"
2 | sort 0 _time
3 | streamstats current=f last(_time) as prevtime by src, dest, dest_port
4 | eval timedelta = _time - prevtime
5 | eventstats avg(timedelta) as avg, count as total by src, dest, dest_port
6 | eval upper=avg*1.1
7 | eval lower=avg*0.9
8 | where timedelta > lower AND timedelta < upper
9 | stats count, values(avg) as TimeInterval by src, dest, dest_port, total
10 | eval prcnt = (count/total)*100
11 | where prcnt > 90 AND total > 10
```

The search results show 76 events (before 7/6/24 2:45:26.000 AM) with no event sampling. The results are displayed in a table with the following columns: src, dest, dest_port, total, count, TimeInterval, and prcnt. The data shows a high percentage of events (93.33333333333333) for the source 10.0.10.20 and destination 192.168.151.181, with a total of 80 events and a count of 15.

src	dest	dest_port	total	count	TimeInterval	prcnt
10.0.10.20	192.168.151.181	80	15	14	60	93.33333333333333

Conclusion

Detecting beaconsing malware is critical for early threat detection and response. By leveraging Splunk's capabilities to monitor network and endpoint logs for anomalous beaconsing patterns, organizations can enhance their security posture by detecting and mitigating potential malware infections before they escalate into serious incidents.